

Lecture 5

Distributions, Normal Model

Part 1.

Distributions

Continuous Random Variables

Let X be a **continuous random variable** taking values between 0 and 10.

there are infinitely many possible values between 0 and 10, so supposing all values are **equally likely** the probability of drawing one specific number is **zero**:

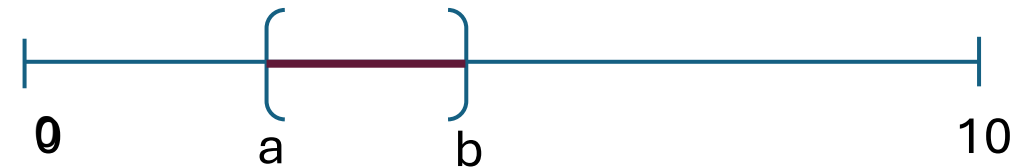
$$P(X = x) = 0, \text{ for any particular value } x.$$

But we can talk about the probability that X falls within an interval.

For example, for an interval (a, b) we can ask: $P(a < X < b)$

If we assume every outcome is equally likely, the only possible response is:

$$P(a < x < b) = \frac{b - a}{10}$$



Probability Density Function

This idea can be generalized.

Suppose we have a continuous function $f(x)$ defined on $[A,B]$.

For $f(x)$ to describe probabilities, it must satisfy two conditions for every value x in the interval $A \leq x \leq B$

- ❑ the function must never be negative: $f(x) \geq 0$
- ❑ the **total area under the curve** must be equal to 1. That is,

$$\int_A^B f(x) dx = 1$$

Such a function is called a **probability density function**, or **PDF**.

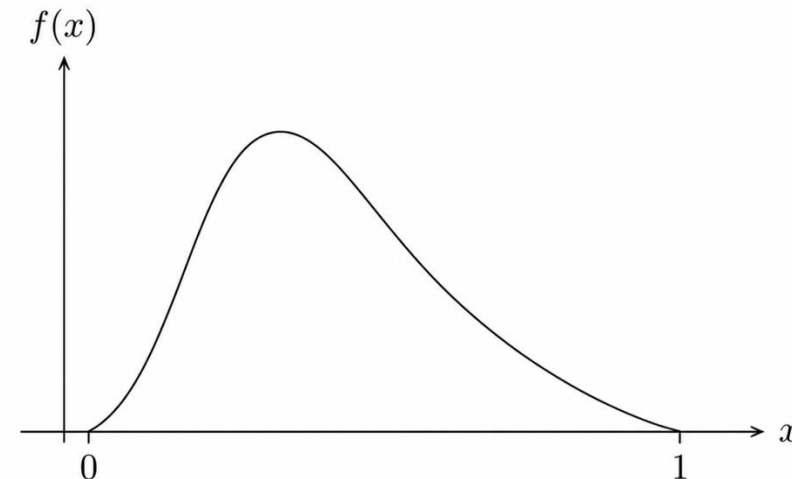
Describing a Continuous Random Variable

In the **discrete** setting, a random variable is described by the finite list of its possible values and by the probability of each of those values

**Probability Distribution
of discrete variable**

Value of X	Prob.
-3	1/8
-1	3/8
1	3/8
3	1/8

For a **continuous** random variable, a continuous random variable is described by the **range** of its possible values and by a **probability density function** $f(x)$.



This curve has the shape of a Beta distribution, but the name is not important. It is simply an example of a possible probability density function.

Uniform distribution

A **uniform distribution** describes a random variable whose values are spread evenly over an interval.

We write:

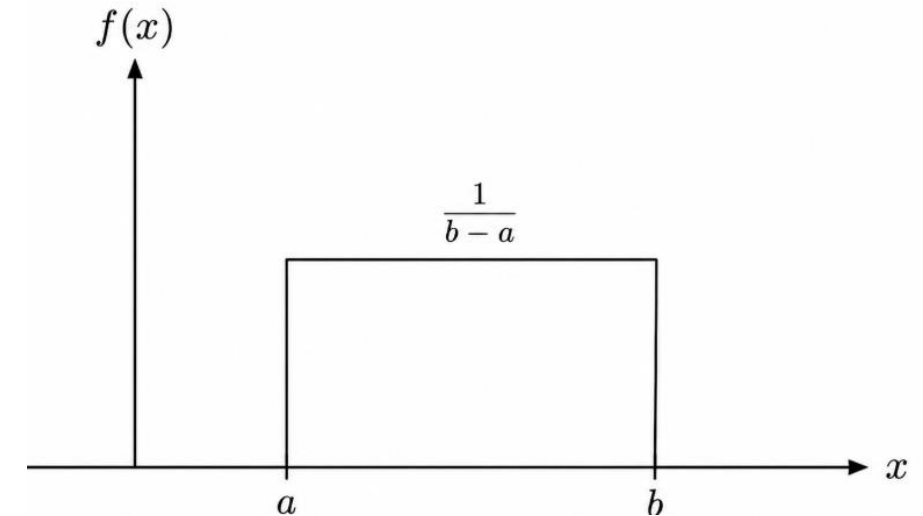
$$X \sim U(a, b)$$

to mean that X can take any value between a and b , with the same density across the whole interval.

The interval $[a, b]$ defines the range of possible values. The probability density function is constant:

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$$

As we saw the probability of falling inside a subinterval depends only on the length of that subinterval.



Exponential distribution

An exponential distribution models the waiting time until the next random event.

$$X \sim \text{Exp}(\lambda)$$

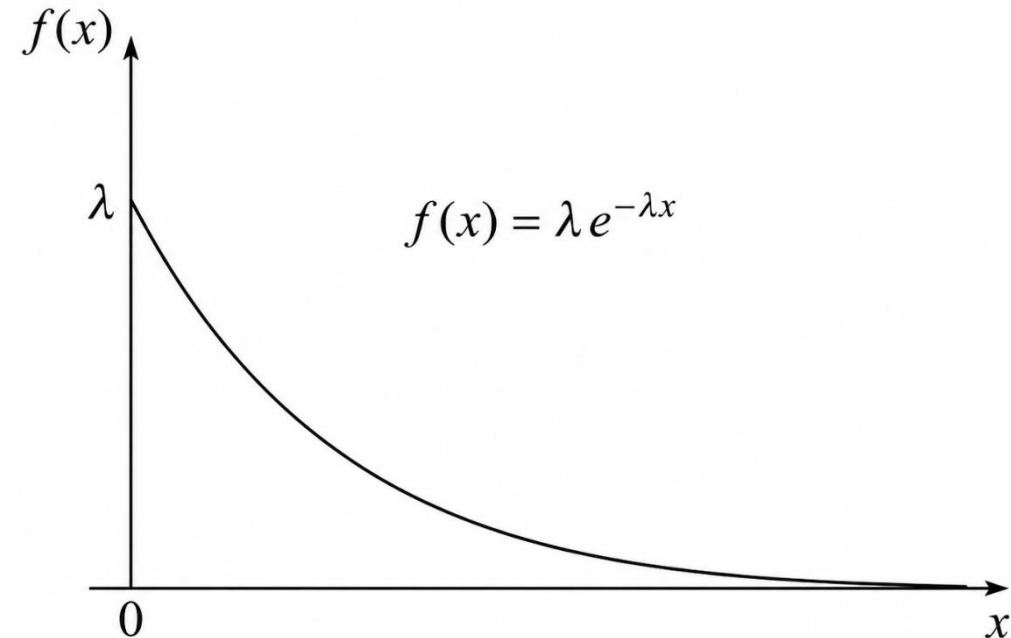
The density is: $f(x) = \lambda e^{-\lambda x}, x \geq 0$

The parameter λ is the **event rate**.

- Large λ : events happen frequently, so waiting times are usually short.
- Small λ : events happen less frequently, so longer waiting times become more likely.

Typical uses:

- waiting time until the next customer arrives,
- time between calls,
- time until a machine failure.



Normal distribution

A normal distribution models values concentrated around a central value, with symmetric variation on both sides.

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

The density is:

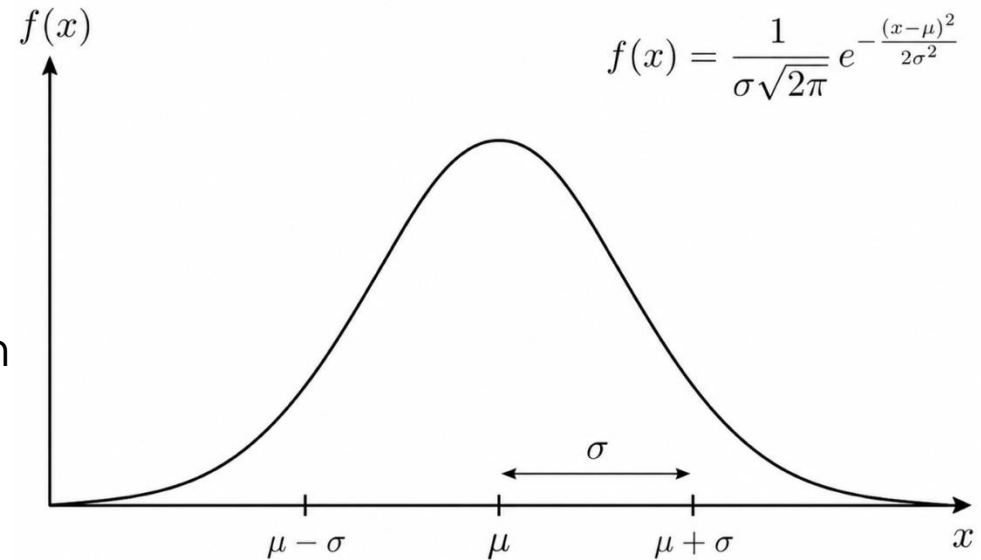
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

The parameter μ determines the center of the distribution.
The parameter σ determines the spread.

- Small σ : values are tightly concentrated around the mean
- Large σ : values are more dispersed.

Typical uses:

- measurement errors,
- returns,
- variables influenced by many small independent effects.



Using a Continuous Distribution in Practice

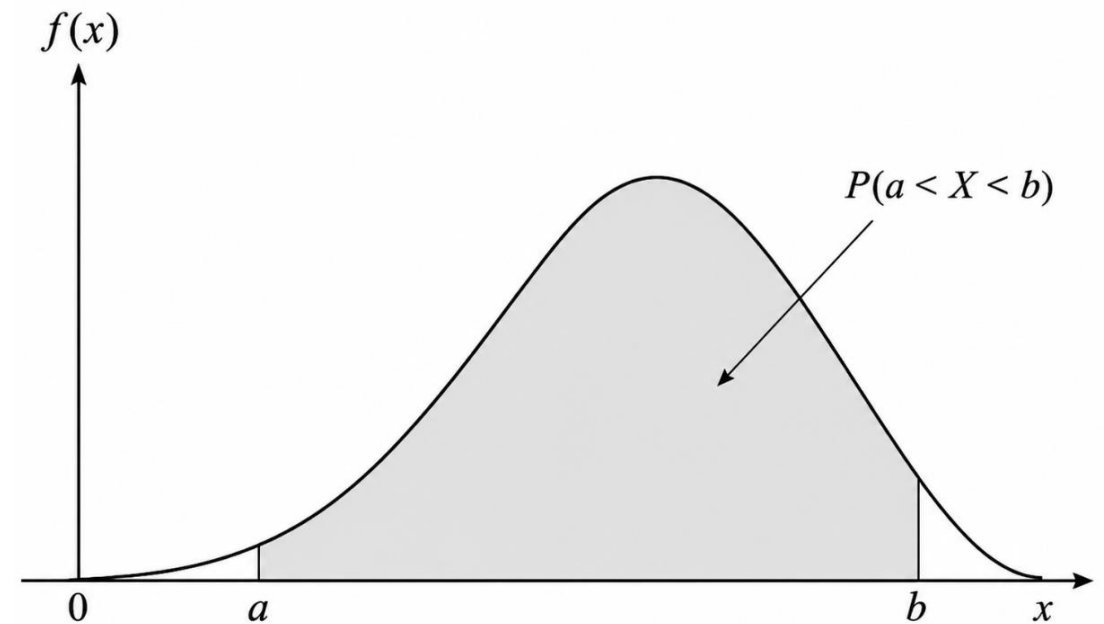
Once we have chosen an appropriate continuous distribution for a problem, we can use it to compute probabilities.

In practice, this means computing the probability that the variable falls within a given interval.

For example: $P(a < X < b)$

This probability is the **area under the probability density curve between a and b**.

For some distributions, this area can be computed analytically. For others, it must be computed numerically.



For the most commonly used distributions, these calculations are already implemented in calculators and statistical software such as Python,